

Linear Regression:

↳ It is one of the easiest and most popular machine learning algorithms.

↳ It is a statistical method that is used for predictive analysis.

↳ Linear regression makes predictions for continuous / real numeric variables.

↳ Linear regression shows a linear relationship between a dependent (Y) and one or more independent variables (X).

↳ Since linear regression shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.

↳ The linear regression model provides a sloped straight line representing the relationship between the variables.

↳

①

Mathematical Understanding of Linear Regression.

In a machine Learning problem, linearity is defined as the linearly dependent nature of a set of independent features x of ~~set of~~ and the dependent quantity y .

(4) Mathematically, if $x = [x_1, x_2, \dots, x_n]$ is a set of independent features and y is a dependent quantity, we try to find a function that maps $y \rightarrow x$ as,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

The β_i 's are the parameters (also called weights) that our linear regression algorithm learns while mapping x to y using supervised historical data.

Ex

Let's understand these mathematical terms via an example. Suppose we want to predict the price of any house. The crucial features that can affect this price are the size of the house, distance from the railway station/airport. Let's treat all these records as a separate feature, then.

x_1 = size of house

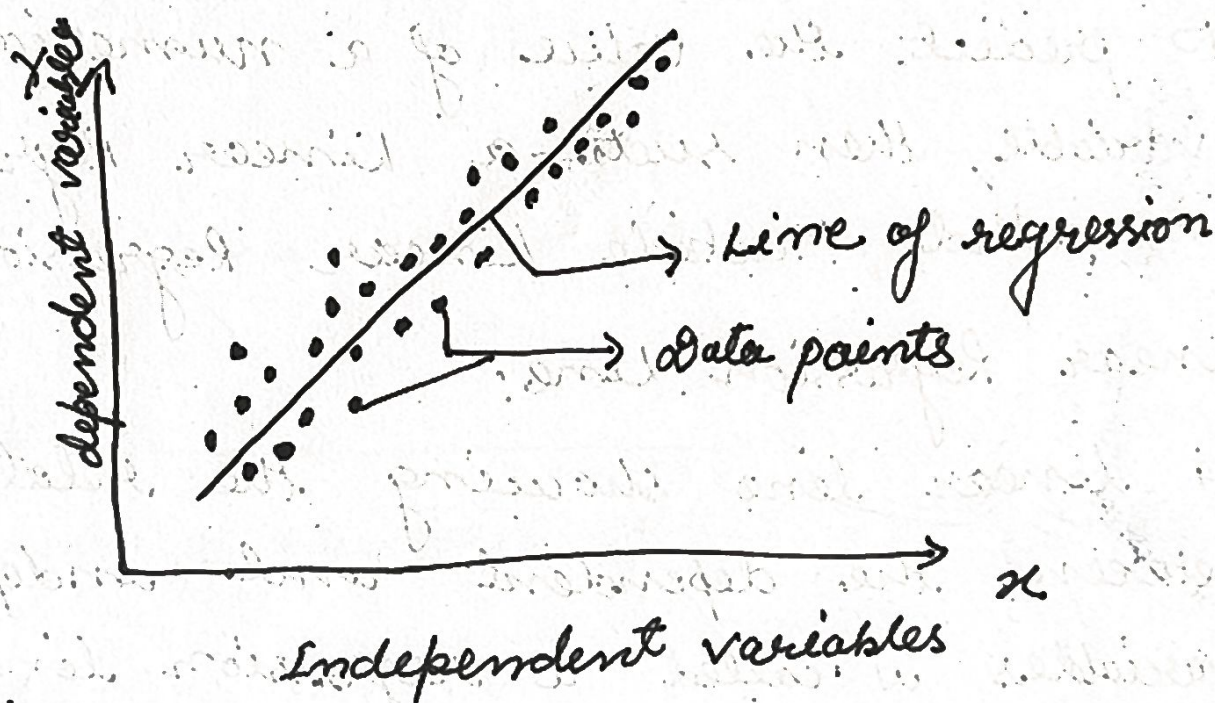
x_2 = distance from airport

x_3 = " " school

→ Every x has a certain weightage like size matters the most and distance from airport matters the least.

→ Hence we need to multiply feature with a real number describing their weightage.

→ Thus β_s are weightage of feature.



Assumptions:

- 1) Linearity: The relationship b/w i/p and o/p is linear
- 2) Independence: Observations are independent to each other (No autocorrelation)
- 3) Normality of Residual: The errors follow a standard normal distribution
- 4) No multicollinearity: Predictor variables are not highly correlated with each other.

Types of Linear Regression

③ Linear regression can be further divided into two types of the algorithm.

1) Simple Linear Regression

If a single independent variable is used to predict the value of a numerical dependent variable, then such a linear regression algo. is called simple linear regression.

2) Multiple Linear Regression

If more than one independent variable is used to predict the value of a numerical dependent variable, then such a linear regression algo. is called Multiple Linear Regression.

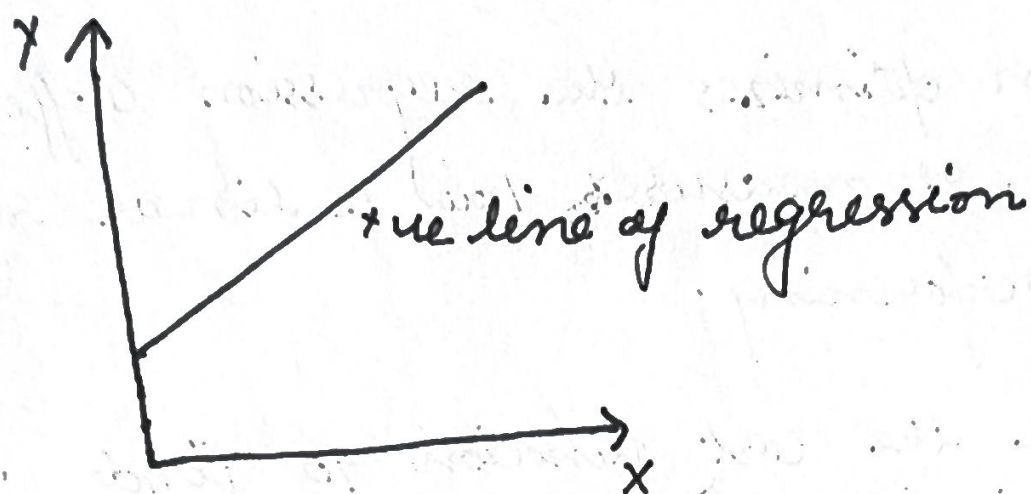
Linear Regression line:

A linear line showing the relationship between the dependent and independent variables is called a regression line. A regression line can show two types of relationship:

1) Positive Linear Relationship:

If the dependent variable increases on the Y-axis and independent variable increases on X-axis, then such a relationship is termed as a positive linear relationship.

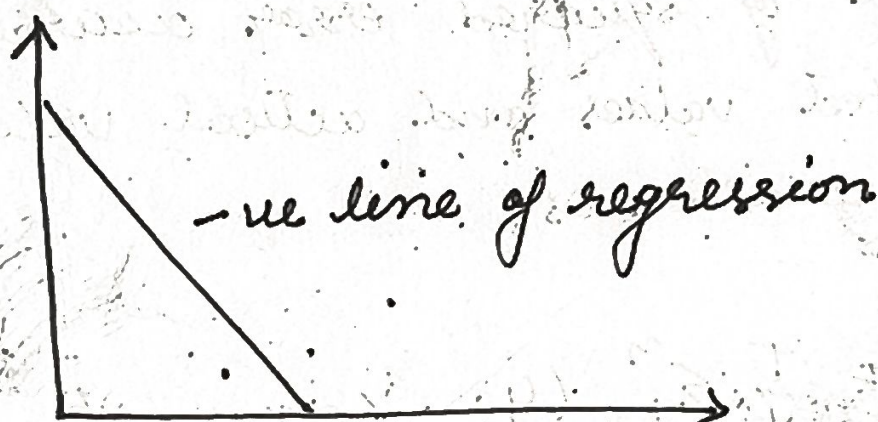
(a)



The line equation will be $y = a_0 + a_1x$

Negative Linear Relationship

If the dependent variable decreases on y-axis and independent variable increases on x-axis, then such a relationship is called a negative linear relationship.



The line of equation will be: $y = -a_0 + a_1x$

Cost function:

↳ The different values for weights or coefficient of lines (a_0, a_1) gives the different line of regression, and the cost function is used to estimate the values of the coefficient for the best fit line.

↳ Cost function optimizes the regression coefficients or weights. It measures how a linear regression model is performing.

↳ We can use the cost function to find the accuracy of the mapping function, which maps the input variable to the output variable. This mapping function is also known as hypothesis function.

⑥ ↳ For Linear Regression, we use the ~~mean~~ Mean Squared Error (MSE) cost function, which is the average of squared error occurred between the predicted values and actual values. It can be written as:

$$\therefore \text{MSE} = \frac{1}{N} \sum_{i=1}^n (y_i - (a_1 x_i + a_0))^2$$

where N = Total no. of observation

y_i = Actual value.

$a_0 + a_1 x_i$ = Predicted value.

Regression model performance evaluation.

↳ There are three error metrics that are commonly used for evaluating and reporting the performance of a regression model.

↳ Mean Square Error (MSE)

↳ Root Mean Squared Error (RMSE)

↳ Mean absolute Error (MAE)

(7)

Mean Square Error (MSE)

↳ It measures the amount of error in statistical model.

↳ Evaluate the mean squared difference b/w observed and predicted values.

↳ If the model has no error, the mean MSE is zero.

↳ Its value increases as the model error increases.

↳ It is always positive value.

↳ It is risk function that corresponds to the predicted squared error loss value.

↳ MSE decreases as the data points are more in line with regression line, indicating less errors in the model.

$$MSE = \frac{\sum (y_i - \hat{y}_i)^2}{n}$$

y_i = is actual value

$\hat{y}_i$ = is the predicted value

n = no. of samples

Root Mean Squared Error (RMSE)

↳ It is most famous equation metric for regression model.

↳ The overall calculation of RMSE is similar to MSE; the final value is square-rooted as we calculated the square of error in MSE.

↳ It is also k/a root mean square deviation

↳ It is one of the most commonly used measures for evaluating the quality of prediction

↳ It shows how far predictions fall from measured true values using Euclidean distance

↳ To compute RMSE, calculate the residual (difference between prediction and truth) for each data point; ~~compute the norm of residual for each~~ compute the mean of residuals and take the square root of that mean.

↳ RMSE is commonly used in supervised learning application.

Root mean square error can be expressed as

$$RMSE = \sqrt{\frac{\sum_{i=1}^N \|y_i - \hat{y}_i\|^2}{N}}$$

→ It is the measure of how well a regression line fits the data points.

→ Using RMSE, we can easily measure the efficiency of the model.

→ If RMSE is measured in the same units as the target variable.

MAE (Mean Absolute Error):

↳ It is a measure of the average size of the mistakes in a collection of predictions, without taking their direction into account.

↳ It is measured as the average absolute difference between the predicted values and the actual values and is used to assess the effectiveness of regression model.

MAE is calculated as the sum of absolute errors divided by the sample size.

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

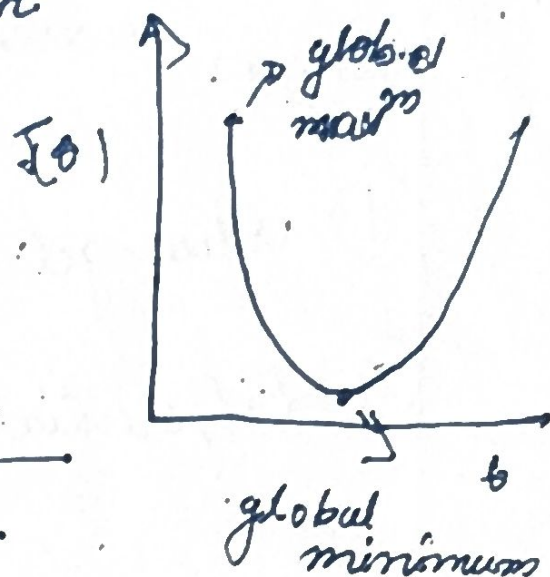
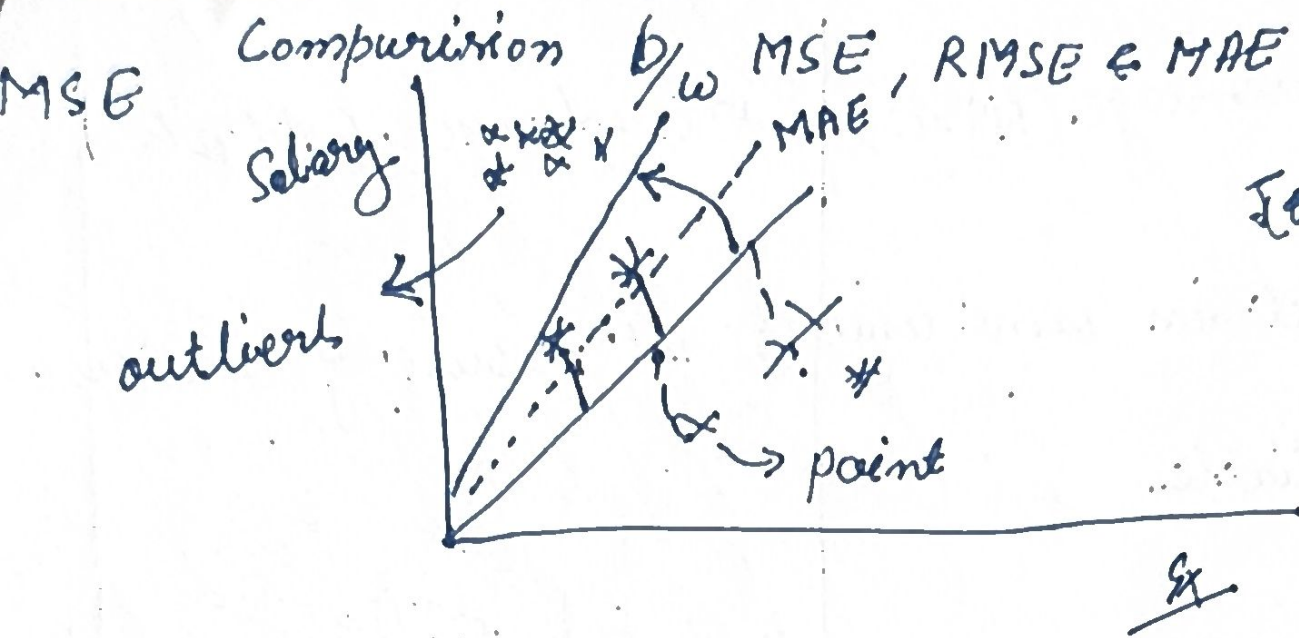
y_i = actual value

$\hat{y}_i$ = Predicted value

n = no. of sample.

→ MAE uses the same scale as the data being measured.

→ This is K/a a scale dependent accuracy measure and therefore cannot be used to make comparisons b/w predicted values that use different scales.



$$MSE = \frac{1}{n} \sum_{i=1}^N (y - \hat{y})^2$$

cost function (error function)

adv. of MSE

- ① Differentiable
- ② It has one local and one global minimum
- ③ It convergence

Disadvantage MSE

- ② Not robust to outliers
- ③ It is no longer in same unit

MAE (advantage)

Disadvantage

- ① Robust to outliers
- ② It will be in same unit

- ① Convergence usually takes more time
- ② Optimisation is a complex task
- ③ It is time consuming

~~MAE~~ advantage (RMSE) Disadvantage (RMSE)

(1) value get in same unit (3) not robust of outliers.

(2) Differentiable.

$$RMSE = \sqrt{MSE}$$

$$= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

→ Gradient decent of MAE



→ It part by divide then
function
i.e. use sub gradient, use
for find global minimum

→

Qx

$$y = B_0 + B_1 x$$

House size	Actual price	$(x_i - \bar{x})$	$(y_i - \bar{y})$
10 (1,000 sq. ft)	200 (\$200k)	$10 - 17.5 = -7.5$	$200 - 290 = -90$
15	260	$15 - 17.5 = -2.5$	$260 - 290 = -30$
20	310	$20 - 17.5 = 2.5$	$310 - 290 = 20$
25	390	$25 - 17.5 = 7.5$	$390 - 290 = 100$

$$\bar{x} = \frac{10 + 15 + 20 + 25}{4} = \frac{70}{4} = 17.5$$

$$\bar{y} = \frac{200 + 260 + 310 + 390}{4} = \frac{1160}{4} = 290$$

$$B_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{-7.5 \times -90 + -2.5 \times -30 + 2.5 \times 20 + 7.5 \times 100}{(-7.5)^2 + (-2.5)^2 + (2.5)^2 + (7.5)^2}$$
$$= \frac{1550}{125} = 12.4$$

$$B_0 = \bar{y} - B_1 \bar{x}$$

$$= 290 - 12.4 \times 17.5$$

$$= 73$$

Final Equation

$$\hat{y} = 73 + 12.4x$$

Now predicting for $x = 18$

$$\hat{y} = 73 + 12.4 \times 18$$

$$= 296.2$$

$$\begin{array}{r} x \quad 10 \quad 200 \quad 60.5^2 \\ 15 \quad 260 \quad 5 \\ 20 \quad 310 \quad 4.80 \\ 25 \quad 390 \quad 32.011 \\ \hline 18 \quad 715.8 \quad 25\% \end{array}$$